

Hishaam Ali Abbasi

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PROFESSIONAL SUMMARY

Robotics engineering graduate with end-to-end experience in mechanical design, embedded firmware, control systems, and software development. Strong track record of taking projects from concept to working prototype across disciplines — from a high-fidelity medical training device to a production home server running containerised services for a family of four. Practical, build-test-iterate approach to engineering with strong leadership credentials. Seeking a graduate engineering role in robotics, embedded systems, or product development.

TECHNICAL SKILLS

Programming Languages: Python (Advanced), Java (Advanced), C++ (Intermediate), SQL (Basic), C (Basic), HTML/CSS/JavaScript (Basic)

Embedded Systems and Control: ESP32, Arduino, Raspberry Pi, ROS2, PID control loops, kinematics, haptics, sensor integration, NVENC hardware transcoding

CAD and Manufacturing: Onshape (Advanced), SolidWorks (Intermediate), Creo (Intermediate), Inventor/Fusion 360 (Intermediate), Ansys FEA (Basic), 3D printing FDM (Advanced), Orca slicer (Advanced), AutoCAD (Basic)

Infrastructure and DevOps: Linux (Debian, Ubuntu), Proxmox VE, Docker, Docker Compose, LXC containers, Nginx, Cloudflare Tunnel, Cloudflare Zero Trust, AWS S3, AWS Glacier, systemd, GPU passthrough, rclone

Software and Tools: Git, MS Office Suite, Adobe Photoshop, Adobe Illustrator, DaVinci Resolve, Portainer, Nextcloud, Duplicati

Languages: Native English, Conversational Urdu

Other: Full UK driving licence

EXPERIENCE

Engineering Design and Research Intern

August 2025 (2 weeks)

Alvis Robotics, Birmingham — EOD (bomb disposal) robot specialists

- Debugged firmware issues and diagnosed mechanical failure points on deployed bomb disposal robot platforms.
- Supported overhaul of legacy robotic systems, identifying components for upgrade and replacement.
- Assisted the Managing Director with R&D planning and financial management tasks, developing well-rounded operational skills.

HSBC Spring Insight Programme — Quantitative Stream

Jul–Aug 2024 (3 weeks)

- Analysed financial datasets and explored quantitative modelling workflows within the investment banking division.
- Gained insight into sector-specific culture and evaluated career paths at the intersection of engineering and finance.

Clinical Data Administrator

Jul–Aug 2021

AT Medics, London

- Processed and managed large volumes of clinical data, improving system efficiency and data governance.
- Eliminated operational backlogs in medical practice management systems.

COVID Clinic Front of House

Dec 2020–Mar 2021

AT Medics, London

- Registered patients and guided non-English speakers through the vaccination process in a high-throughput clinical environment.
- Developed strong communication and interpersonal skills working with diverse patient groups.

PROJECTS

Home Server and Self-Hosted Infrastructure

2025–Present

- Designed and built a complete home server on a Lenovo ThinkStation (Xeon E3-1230v5, 16GB RAM, 9TB storage, RTX 3050) running Proxmox VE as a bare metal hypervisor.
- Configured Nvidia GPU passthrough to an LXC container for NVENC hardware transcoding, resolving kernel pinning, cgroup device passthrough, and driver version mismatch challenges.
- Deployed containerised services via Docker Compose: Jellyfin (4K media with HDR tone mapping), Nextcloud (8TB family cloud for 4 users with CardDAV/CalDAV sync), Nginx Proxy Manager, Portainer, and Duplicati.
- Automated encrypted nightly backups to AWS S3 Glacier Deep Archive with AES-256 encryption and smart retention policies.
- Configured Cloudflare Tunnel and Zero Trust Access across 5 custom domains, eliminating open ports. Migrated 4 websites from Vercel to self-hosted Nginx containers.

- Resolved complex technical challenges including AppArmor conflicts, systemd auto-creation of Nvidia UVM devices, and PHP timeout tuning for Nextcloud.

Fracture Immobilisation Trainer — Final Year Dissertation

2025–2026

- Led development of a high-fidelity cast application trainer to bridge the gap between theory and clinical practice for medical trainees.
- Engineered a fluidic pump architecture simulating subcutaneous tissue response under cast pressure.
- Integrated force sensors with an ESP32 microcontroller to calculate a real-time pain index, providing immediate trainee feedback.
- Combined rapid prototyping, embedded systems, haptics, and sensor integration into a single clinical training device.

Formula Student — Suspension Team

Oct 2023–Present

Queen Mary Racing

- Optimised suspension geometry using Ansys FEA to balance grip, stiffness, and durability for improved track performance.
- Contributed to the full redesign of the suspension system for a new car platform.
- Collaborated cross-department (powertrain, aerodynamics, chassis) to deliver a cohesive high-performance vehicle design.

Active Camera Gimbal

- Built a 2-axis gyroscopically stabilised camera gimbal with custom C++ firmware and real-time PID control loops on an embedded microcontroller.

BoxTurtle Multi-Material 3D Printing System

- Automated filament swaps on a modified Ender 3 with hardware modifications and custom macros, enabling reliable multi-material printing workflows.

Additional Projects

Robotic arm, web development (bones-of-steel.com), e-commerce cookie cutter shop, ongoing product development. Full portfolio with detailed case studies at hishaamabbasi.co.uk.

EDUCATION

BEng Robotics Engineering

2022–2026

Queen Mary, University of London

- Key modules: Numerical Methods and Data Science (86%), Control Systems and Analysis (78.9%), Modelling and Control of Robotic Systems (75.2%), Design for Human Interaction (70.2%).
- Dissertation: High-fidelity fracture immobilisation trainer with fluidic pump architecture, haptic feedback, and real-time ESP32 sensing.

Sutton Grammar School

2015–2022

- A-Levels: Mathematics (A), Physics (A), Computer Science (B).
- 12 GCSEs including Mathematics (9), Computer Science (8), Physics (8).

LEADERSHIP

Explorer Scout Leader

2012–Present

- Achieved the Chief Scout Diamond Award, the highest level in Scouting.
- Currently plan and deliver weekly sessions, expeditions, and residential camps for Explorer Scouts.
- Founded a new Explorer section to serve older Scouts in the district.

Combined Cadet Force — Flight Sergeant

Sep 2018–Jul 2023

- Progressed to the highest NCO rank; taught the CCF syllabus, led teams on camps, and mentored younger cadets.
- Developed adaptive leadership skills across varied environments and group compositions.